

Supplementary File for: Consideration of Control-Loop Interaction in Transient Stability of Grid-Following Inverters using Bandwidth Separation Method

APPENDIX A

FULL-ORDER ODE MODEL OF GFL INVERTER

In this paper, the PLL is designed to be over-damped, while the DVC and the inner current loop are implemented as typical second-order systems.

The PI controllers are designed in the small-signal domain. The closed-loop transfer function of the PLL is given by

$$G_{\text{pll}}(s) = \frac{\Delta\delta}{\Delta V_q} = \frac{k_{\text{p,pll}} + k_{\text{i,pll}}/s}{s + V_d(k_{\text{p,pll}} + k_{\text{i,pll}}/s)}$$

according to [1]. Assuming $V_d \approx 1$ at steady state and neglecting the integral term, which corresponds to an overdamped PLL design, we have $G_{\text{pll}}(j\omega_{\text{pll}})|_{\omega_{\text{pll}}=k_{\text{p,pll}}} = -3$ dB. This overdamped setting is commonly adopted since the small integral term and large proportional term are beneficial to transient stability [2]–[4]. In this paper, we define $\zeta_{\text{pll}} = k_{\text{i,pll}}/k_{\text{p,pll}}$.

The closed-loop transfer function of the DVC is:

$$G_{\text{dvc}}(s) = \frac{k_{\text{p,dvc}} + k_{\text{i,dvc}}/s}{sC_{\text{dc}}/\omega_s + k_{\text{p,dvc}} + k_{\text{i,dvc}}/s}$$

and the gains are selected as $k_{\text{p,dvc}} = \frac{C_{\text{dc}}/\omega_s}{2} \cdot \omega_{\text{dvc}}$ and $k_{\text{i,dvc}} = \zeta_{\text{dvc}} \frac{C_{\text{dc}}/\omega_s}{2} \cdot (\omega_{\text{dvc}})^2$. When $\zeta_{\text{dvc}} = 1/4$, the DVC becomes a critically damped second-order system [5], with $G_{\text{dvc}}(j\omega_{\text{dvc}}) \approx -3$ dB. The same PI design approach is also applied to the inner current loop.

By substituting the above expressions for loop bandwidths, the nonlinear differential equations for PLL, DVC, and TVC, as Eq. (1), (2), and (3), can be reformulated as Eq. (4) in the paper, and the detailed expressions in Eq. (4) are as follows:

$$\begin{aligned} f_{\text{pll},1}(\delta, i_d, \frac{x_{\text{int,pll}}}{\omega_{\text{pll}}}) &= \frac{X_g i_d - U_g \sin \delta + x_{\text{int,pll}}/\omega_{\text{pll}}}{1 - k_{\text{p,pll}} L_g / \omega_s \cdot i_d} \\ f_{\text{pll},2}(\delta, i_d, x_{\text{int,pll}}) &= \zeta_{\text{pll}} \frac{X_g i_d - U_g \sin \delta + x_{\text{int,pll}} L_g / \omega_s}{1 - k_{\text{p,pll}} L_g / \omega_s \cdot i_d} \end{aligned} \quad (\text{A1a})$$

$$\begin{aligned} f_{\text{dvc},1}(\delta, i_{\text{dq}}, \omega_{\text{dvc}} \Delta v_{\text{dc}}) &= P_{\text{in}} - i_d U_g \cos \delta + i_q U_g \sin \delta \\ &\quad + \frac{C_{\text{dc}}/\omega_s}{2} \zeta_{\text{dvc}} \omega_{\text{dvc}} \Delta(v_{\text{dc}})^2 \\ f_{\text{dvc},2}(\delta, i_{\text{dq}}) &= \frac{2}{C_{\text{dc}}/\omega_s} (P_{\text{in}} - i_d U_g \cos \delta \\ &\quad + i_q U_g \sin \delta) \end{aligned} \quad (\text{A1b})$$

$$f_{\text{tvc}}(\delta, i_q) = -i_q + k_v U_g \cos \delta - k_v i_q X_g - k_v V_{\text{ref}} \quad (\text{A1c})$$

APPENDIX B

DETAILED REDUCED-ORDER MODEL DERIVATION

A. Bandwidth sequence: ω_{pll} larger than ω_{dvc}

The PLL represented by Eq. (1) is considered as fast subsystem, and defining $\varepsilon = \frac{1}{\omega_{\text{pll}}} \cdot (1 - k_{\text{p,pll}} L_g / \omega_s \cdot i_d)$, It

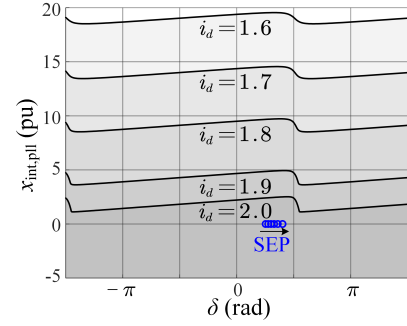


Fig. A1. Stability region of PLL under different i_d ($\omega_{\text{pll}} = 15 \text{ Hz} \times 2\pi$).

clear that as $\omega_{\text{pll}} \rightarrow \infty$, $\varepsilon \rightarrow 0$. Introducing the fast time-scale $\tau = t/\varepsilon$, the fast subsystem equations become:

$$\begin{cases} \frac{d}{d\tau} \delta = (X_g i_d - U_g \sin \delta) + \frac{x_{\text{int,pll}}}{\omega_{\text{pll}}} \\ \frac{d}{d\tau} x_{\text{int,pll}} = \zeta_{\text{pll}} (X_g i_d - U_g \sin \delta + x_{\text{int,pll}} L_g / \omega_s \cdot i_d) \end{cases} \quad (\text{A2})$$

Since ω_{pll} is large, $x_{\text{int,pll}}/\omega_{\text{pll}} \approx 0$. Under a fixed i_d , this subsystem rapidly converges to a SEP defined by:

$$\bar{\delta} = \text{asin} \left(\frac{X_g i_d}{U_g} \right), \bar{x}_{\text{int,pll}} = 0 \quad (\text{A3})$$

The SEP is uniformly exponentially stable if the following conditions hold:

$$i_d < U_g / X_g, \zeta_{\text{pll}} < \frac{\sqrt{U_g^2 - (X_g i_d)^2}}{i_d L_g / \omega_s}, i_d < 1/\omega_{\text{pll}} L_g \quad (\text{A4})$$

In practice, the second and third conditions are generally nonrestrictive. Since the PLL bandwidth is typically much smaller than the fundamental frequency ($L_g \omega_{\text{pll}} < X_g$), the third condition is readily satisfied. Moreover, an overdamped PLL ($\zeta_{\text{pll}} < 1$) is adopted in this paper, and the right-hand side of the second inequality is typically much larger than unity. Therefore, only the dominant practical condition, $i_d < U_g / X_g$, is retained in the main text for clarity. In addition, the stability region for different i_d is illustrated in Fig. A1 under $\zeta_{\text{pll}} = 1/4$. Following singular perturbation theory, we represent the fast dynamics as:

$$\delta(t) = \bar{\delta}(t) + \delta^*(\tau) + O(\varepsilon) \quad (\text{A5})$$

where the $\delta^*(\tau)$ describes the rapid convergence of the fast subsystem immediately after the disturbance and satisfies:

$$\frac{d}{d\tau} \delta^*(\tau) = X_g i_d - U_g \sin \left(\text{asin} \left(\frac{X_g i_d}{U_g} \right) + \delta^*(\tau) \right) \quad (\text{A6})$$

The DVC loop is the slow subsystem. Following the asymptotic expansion methodology [6], [7], the slow variables are expanded as:

$$i_d = \bar{i}_d + \varepsilon i_d^*(\tau) + O(\varepsilon) \quad (\text{A7})$$

where $\bar{i}_d$ and $\Delta(v_{\text{dc}})^2$ satisfy the reduced-order slow-subsystem differential equations:

$$\frac{d}{dt} \begin{bmatrix} \bar{i}_d \\ \Delta(v_{\text{dc}})^2 \end{bmatrix} = \begin{bmatrix} \omega_{\text{dvc}} f_{\text{dvc},1}(\bar{\delta}, i_{\text{dq}}, \omega_{\text{dvc}} \Delta(v_{\text{dc}})^2) \\ f_{\text{dvc},2}(\bar{\delta}, i_{\text{dq}}) \end{bmatrix} \quad (\text{A8})$$

with $\bar{\delta} = \text{asin} \left(\frac{X_g i_d}{U_g} \right)$ substituted from the equilibrium of the fast subsystem. This resulting reduced-order slow-subsystem model is identical to Eq. (7b) in the main text.

The boundary-layer correction terms $\varepsilon \dot{i}_d^*(\tau)$ and $\varepsilon [\Delta(v_{dc})^2]^*$ represent the small influence of the rapid initial convergence of the fast PLL subsystem on the slow DVC subsystem. They are of order $O(\varepsilon)$ and rapidly decay to zero. Therefore, they do not alter the reduced-order slow-subsystem differential equations, namely (A8) (Eq. (7b) in the main text). Accordingly, these correction terms can be neglected in the reduced-order model, yielding the standard first-order singular-perturbation approximation. If higher-order approximation accuracy is desired, this influence can be equivalently incorporated as small corrections to the initial values of the slow variables. Consequently, the reduced-order model retains the same differential equations but starts from the corrected initial values, improving the overall approximation accuracy from $O(\varepsilon)$ to $O(\varepsilon^2)$. Accordingly, the corrected initial conditions are $i_d(0) - i_d^*(0)$ and $\Delta(v_{dc})^2(0) - [\Delta(v_{dc})^2]^*(0)$, where the corresponding analytical expressions are given by:

$$\begin{aligned} i_d^*(0) = & -\omega_{dvc} \int_0^\infty f_{dvc,1}(\bar{\delta}(0) + \delta^*(\tau), \\ & i_{dq}(0), \omega_{dvc} \Delta(v_{dc})^2(0)) \\ & - f_{dvc,1}(\bar{\delta}(0), i_{dq}(0), \omega_{dvc} \Delta(v_{dc})^2(0)) d\tau \\ & \approx k_{p,dvc} \frac{2}{C_{dc}} U_g i_d(0) \sin \bar{\delta}(0) \cdot \frac{\delta(0) - \bar{\delta}(0)}{U_g \cos \bar{\delta}(0)}. \end{aligned} \quad (A9a)$$

$$\begin{aligned} [\Delta(v_{dc})^2]^*(\tau) = & - \int_\tau^\infty f_{dvc,2}(\bar{\delta}(0) + \delta^*(\tau), \\ & i_{dq}(0), \omega_{dvc} \Delta(v_{dc})^2(0)) \\ & - f_{dvc,2}(\bar{\delta}(0), i_{dq}(0), \omega_{dvc} \Delta(v_{dc})^2(0)) d\tau \\ & \approx \frac{2}{C_{dc}} U_g i_d(0) \sin \bar{\delta}(0) \cdot \frac{\delta(0) - \bar{\delta}(0)}{U_g \cos \bar{\delta}(0)}. \end{aligned} \quad (A9b)$$

where $\delta(0)$ denotes the initial post-fault value, differing from $\bar{\delta}(0)$ which determined by the initial value of slow variables.

B. Bandwidth sequence: ω_{dvc} larger than ω_{pll}

The DVC represented by Eq. (2) is considered as fast subsystem, and defining $\varepsilon = 1/\omega_{dvc}$. It clear that as $\omega_{dvc} \rightarrow \infty$, $\varepsilon \rightarrow 0$. Introducing the fast time-scale $\tau = t/\varepsilon$, and do variable substitution $\sigma = \omega_{dvc} \cdot \Delta(v_{dc})^2$, the fast subsystem equations become:

$$\begin{cases} \frac{d}{d\tau} i_d = (P_{in} - i_d U_g \cos \delta + i_q U_g \sin \delta) + \frac{C_{dc}/\omega_s}{2} \zeta_{dvc} \sigma \\ \frac{d}{d\tau} \sigma = \frac{2}{C_{dc}} (P_{in} - i_d U_g \cos \delta + i_q U_g \sin \delta) \end{cases} \quad (A10)$$

It is a linear differential equation for fixed δ , and it is easy to prove that if $\cos \delta > 0$, the Jacobian is negative, which means that the fast subsystem is globally asymptotical stable. They rapidly converges to $\bar{i}_d$ and $\bar{\sigma}$:

$$P_{in} - \bar{i}_d U_g \cos \delta + i_q U_g \sin \delta = 0, \bar{\sigma} = 0 \quad (A11)$$

Similarly, following singular perturbation theory, the fast-subsystem dynamics can be expressed as:

$$\begin{cases} i_d(t) = \bar{i}_d(t) + i_d^*(\tau) + O(\varepsilon) \\ \sigma(t) = \bar{\sigma}(t) + \sigma^*(\tau) + O(\varepsilon) \end{cases} \quad (A12)$$

where the $i_d^*(\tau)$, $\sigma^*(\tau)$ describes the rapid convergence of the fast subsystem immediately after the disturbance and satisfies:

$$\begin{cases} \frac{d}{d\tau} i_d^*(\tau) = -i_d^*(\tau) U_g \cos \delta(0) + \frac{C_{dc}/\omega_s}{2} \zeta_{dvc} \sigma^*(\tau) \\ \frac{d}{d\tau} \sigma^*(\tau) = -\frac{2}{C_{dc}} i_d^*(\tau) U_g \cos \delta(0) \end{cases} \quad (A13)$$

The PLL loop is the slow subsystem. Following the asymptotic expansion methodology [6], [7] the slow variables are expanded as:

$$\begin{cases} \delta(t) = \bar{\delta}(t) + \varepsilon \delta^*(\tau) + O(\varepsilon) \\ x_{int,pll}(t) = \bar{x}_{int,pll}(t) + \varepsilon x_{int,pll}^*(\tau) + O(\varepsilon) \end{cases} \quad (A14)$$

where $\bar{\delta}(t)$ and $\bar{x}_{int,pll}$ satisfy the reduced-order slow subsystem equations:

$$\begin{bmatrix} \dot{\delta} \\ \dot{x}_{int,pll} \end{bmatrix} = \begin{bmatrix} \omega_{pll} f_{pll,1} \left(\delta, \bar{i}_d, \frac{x_{int,pll}}{\omega_{pll}} \right) \\ \omega_{pll} f_{pll,2} \left(\delta, \bar{i}_d, x_{int,pll} \right) \end{bmatrix} \quad (A15)$$

with $\bar{i}_d$ obtained from the equilibrium condition of the fast subsystem. After simplification, (A15) becomes identical to Eq. (8b) in the main text.

The boundary-layer correction terms $\varepsilon \delta^*(\tau)$ and $\varepsilon x_{int,pll}^*(\tau)$ are $O(\varepsilon)$ terms representing the small effect of the rapid initial convergence of the fast subsystem toward its equilibrium. Since they decay rapidly to zero, they do not alter the reduced-order slow-subsystem differential equations. Consequently, the reduced-order model in (A15) can be directly used without these correction terms, yielding the standard first-order singular-perturbation approximation. If higher-order approximation accuracy is desired, the influence of the rapid convergence of the boundary-layer dynamics on the slow subsystem can be equivalently incorporated as small corrections to the initial values of the slow variables. In this case, the corrected initial conditions for δ and $x_{int,pll}$ become $\delta(0) - \delta^*(0)$ and $x_{int,pll}(0) - x_{int,pll}^*(0)$, respectively.

$$\begin{aligned} \delta^*(0) = & -\omega_{pll} \int_0^\infty f_{pll,1} \left(\delta(0), i_d^*, \frac{x_{int,pll}(0)}{\omega_{pll}} \right) d\tau \\ x_{int,pll}^*(0) = & -\omega_{pll} \int_0^\infty f_{pll,2} \left(\delta(0), i_d^*, x_{int,pll}(0) \right) d\tau \end{aligned} \quad (A16)$$

C. Stability analysis of TVC subsystem

Since the DVC loop does not affect the TVC loop, only the influence of the PLL is considered. The differential equation (3), which governs i_q , is a linear system with δ serving as the input. It is globally exponentially stable uniformly in δ , and therefore input-to-state stable (ISS) according to Lemma 4.6 in [8]. In particular, if $\delta(t)$ converges to a constant, then $i_q(t)$ asymptotically converges to the corresponding steady-state value. In summary, if δ can be stable, then Eq. (3) in the main text is stable.

APPENDIX C

DETAILED SYSTEM PARAMETERS

If not specified, the system parameters used in this paper are listed in Table A1.

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TABLE A1
PARAMETERS IN SIMULATION AND HIL EXPERIMENT

Parameters	Value ¹	
	Simulation	HIL Experiment
Base capacity S_b	1	1 MVA
Base voltage V_b	1	690 V
Base current I_b	1	1.45 kA
Fundamental frequency $\omega_s/2\pi$	50 Hz	50 Hz
Grid voltage U_g	1 pu	690 V
Input power P_{in}	1 pu	1 MW
Short circuit ratio (SCR)	2.1	2.1
Grid impedance Z_g	$0.47j + 0.002$ pu	$0.72\text{mH} + 0.95\text{m}\Omega$
DC voltage reference $V_{dc,ref}$	2.5 pu	1725 V
DC capacitor C_{dc}	12.5 pu	83 mF
DC voltage limiter $V_{dc,limit}$	2-3 pu ($\pm 20\%$)	1380 - 2070 V
PLL PI controller	$k_{p,pll} = \omega_{pll}$ $k_{i,pll} = \zeta_{pll} \cdot k_{p,pll} \cdot \zeta_{pll} = 1/4$	same
DVC PI controller	$k_{p,dvc} = \omega_{dvc} \cdot C_{dc}/2\omega_s$ $k_{i,dvc} = k_{p,dvc} \cdot \omega_{dvc}/4$	same
TVC droop coefficient k_v	2 pu	same
TVC voltage reference $V_{t,ref}$	0.81 pu	560 V
Inner current control loop bandwidth	1 kHz	same
Current limit ² I_{limit}	2.5 pu	3.1 kA
LC filter L_f	0.05 pu	75.7 μH
LC filter C_f	0.01 pu	66.8 μF
Switching frequency	averaging model	10 kHz
DSP control frequency	—	10 kHz

¹ The pu values in simulation and experiment are the same.

² Rectangular current limiter.

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